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                          KURULUM TALIMATLARI
        Yapay Zeka Destekli Harcama ve Fatura Analiz Sistemi
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Versiyon   : 1.0
Tarih      : Mayis 2026
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1. SISTEM MIMARISI VE BILESENLER
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Sistem 4 ana bilesenden olusmaktadir:

1) Mobil Uygulama  : Flutter 3.x (Android / iOS / Web / Masaustu)
2) Backend API     : Java 21 & Spring Boot 3.3.5 (Port: 8080, Path: /api)
3) AI Mikroservisi : Python 3.11 & FastAPI (Port: 8000)
4) Veritabani      : PostgreSQL 16 (Port: 5432) / H2 In-Memory (Gelistirme)

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2. DOCKER COMPOSE ILE HIZLI KURULUM (ONERILEN)
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Tum arka plan servislerini (PostgreSQL + FastAPI AI + Spring Boot API)
tek komutla baslatmak icin:

  cd "3.SOURCE CODES/Source_Code"
  export GEMINI_API_KEY="SIZIN_GEMINI_API_ANAHTARINIZ"
  docker compose up --build -d

Servisleri durdurmak icin:
  docker compose down

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3. KAYNAK KODDAN MANUEL GELISTIRICI KURULUMU
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3.1 Gereksinimler
- Java JDK 21+
- Python 3.10+
- Flutter SDK 3.5+
- PostgreSQL 14+ (veya varsayilan H2 veritabani)
- Google Gemini API Anahtari (Opsiyonel - yoksa EasyOCR devreye girer)

3.2 AI Mikroservisi Kurulumu (Port 8000)
  cd "3.SOURCE CODES/Source_Code/ai-service"
  python -m venv venv
  source venv/bin/activate        # Windows: .\venv\Scripts\activate
  pip install -r requirements.txt
  cp .env.example .env            # GEMINI_API_KEY degerini girin
  python main.py

3.3 Backend API Kurulumu (Port 8080)
  cd "3.SOURCE CODES/Source_Code/backend-api"
  ./mvnw clean spring-boot:run    # Windows: .\mvnw.cmd spring-boot:run

3.4 Mobil Uygulama Kurulumu (Flutter)
  cd "3.SOURCE CODES/Source_Code/mobile_app"
  flutter pub get
  flutter run

Yeni Android APK derlemek icin:
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  flutter build apk --release

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4. SERVIS SAGLIK KONTROLLERI (HEALTH CHECK)
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Servisler baslatildiktan sonra yerel erisim noktalarini dogrulayin:
- Backend API Health : GET http://localhost:8080/api/health
- Backend Swagger UI : GET http://localhost:8080/api/swagger-ui.html
- AI Service Health  : GET http://localhost:8000/health
- AI Service OpenAPI : GET http://localhost:8000/docs
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